METRIC
DOD-L-24681(SH)
12 January 1987

MILITARY SPECIFICATION

LOUDSPEAKER, CONTROL OF VOLUME, AUTOMATIC (METRIC)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, test, manufacture, and acceptance requirements for the Automatic Loudspeaker Control (ALC) for use in areas with large ambient noise level variation (not for application on Nuclear Powered Ships).

1.2 Classification. ALCs shall be of two types, as specified (see 6.2.1).

Type	Application
CA	Central amplifiers - 70.7 volt announcing systems
DA	Distributed amplifiers - 12.2 volt announcing systems

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbook. The following specifications, standards, and handbook form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-16400 - Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore: General Specification for.
MIL-E-17555 - Electronic and Electrical Equipment Accessories and Provisioned Items (Repair Parts): Packaging of.
MIL-S-19500 - Semiconductor Devices, General Specification for.
MIL-M-38510 - Microcircuits, General Specification for.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-471 - Maintainability Verification/Demonstration/Evaluation.
MIL-STD-781 - Reliability Design Qualification and Product Acceptance Tests: Exponential Distribution.
DOD-STD-1399, Section 300 - Interface Standard for Shipboard Systems, Electric Power, Alternating Current. (Metric)

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

2.1.2 Other Government drawings and publications. The following other Government drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

TECHNICAL MANUAL

DEPARTMENT OF THE NAVY

SE101-AA-MMO-101 - Automatic Loudspeaker Controls C-10756/SIC and C-10757/WIC.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

53711-5379400 - Assembly, Automatic Loudspeaker Control, C-10756/SIC.
53711-5379408 - Assembly, Automatic Loudspeaker Control, C-10757/WIC.

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PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
0967-LP-597-1011 - Electronic Equipment, Parts Application and Reliability Information for Navy.

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITIUE (ANSI)
S1.4 - Specification for General Purpose Sound Level Meters.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Characteristics. The purpose of the ALC is to automatically maintain the announcing system at a fixed sound level difference above the ambient background noise. This must be accomplished by attenuating the announcing system distributed signal in accordance with table I. The "Rated" audio signal voltage levels that are distributed are as designated in 1.2. The ALC characteristics shall be as follows:

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TABLE I. Output attenuation/ambient noise levels (see 3.2.4).

Announcing system Signal attenuation dB	Ambient noise dBC
0	100
-6	94-100
-12	88-94
-18	82-88
-24	76-82
-30	76

3.2.1 Operating modes. The ALC shall have three operating modes, designated "Run," "Test," and "Bypass."

3.2.1.1 Run mode. The noise sensing circuitry shall operate continously; however, the ALC signal output level switching and noise level latching shall not occur until an audio signal is sensed (activated condition). The signal onset time shall not exceed 12 milliseconds (see table II). The noise attack time, noise release time, signal release time, and millivolt level for maximum noise level (zero attenuation) shall be as specified in table II.

TABLE II. Signal and noise requirements (see 3.2.1.1).

Characteristics	Requirements
Signal sensing	
(a) Signal onset time	12 ms maximum
(b) Signal release time	8 to 12 seconds
(c) Minimum signal for activation	1.0 V rms sine wave (nominal) 1.3 V rms sine wave (maximum)
Noise sensing	
(a) Noise attack time (to maximum noise level)	6 ± 2 seconds
(b) Noise release time (from maximum noise)	15 ± 2 seconds
(c) Noise input sensitivity	45.0 mV rms maximum for maximum volume (nominal) 90.0 mV rms (-6dB input sensitivity) 22.5 mV rms (+6dB input sensitivity)

3.2.1.1.1 Monitoring circuits. If the noise level in the deactivated state drops 40 decibels (dBC) below the set level for zero volume attenuation and stays below this level throughout a subsequent activation (audio signal) and deactivation, then the ALC shall automatically test its noise sensing circuits one time after each activation. If the built-in test equipment (BITE) indicates external circuit (to the microphone) impedance change or abnormally low voltage for the

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test signal, then the ALC shall go into Bypass mode and start a failure light flashing. If the ALC direct current (dc) power supply voltages become abnormal, the ALC shall go into Bypass mode and turn off its power source input. If the ALC dc power supply voltages become abnormal, the ALC shall go into BYPASS mode and turn its power source input off.

3.2.1.2 Test mode. When in Test mode, the ALC shall also be in Bypass mode. BITE shall be provided to test the noise sensing circuitry, internal and external, and to indicate and set up the signal output attenuator levels. A built-in signal generator shall be supplied for noise and signal input simulation. The ALC output attenuator setting shall be indicated. The test signal level shall also be indicated.

3.2.1.3 Bypass mode. The ALC shall revert to the Bypass mode from the RUN mode and from OFF automatically for any of the following:

(a) Bypass mode selection.
(b) Dc supply failure or partial failure.
(c) Noise circuit failure (see RUN mode).
(d) Test mode selection.

3.2.2 Wiring. Signal (noise or audio) wiring shall be electrically insulated from chassis ground. The chassis shall provide for connection of grounding cable.

3.2.3 Noise input circuits. The noise sensing, amplification, filtering, and detection shall be in accordance with ANSI S1.4 for type 3 sound level meters having "C" weighting only. A standard meter movement with vibration isolation shall be provided. The meter movement shall be replaceable in less than 10 minutes without special tools.

3.2.3.1 Microphone. The microphone performance shall be as specified in ANSI S1.4 to sense dBC noise levels and to provide for usual sound level meter calibration methods.

3.2.3.2 Microphone mounting. Vibration isolation mounting to bulkhead or ceiling is required. Calibration with and without removal from the mounting is required. The microphone shall be replaceable without special tools in less than 3 minutes.

3.2.3.3 Microphone signal mixing. Application of two microphones to better sense compartment ambient noise levels shall be provided for and full calibration and installation instructions given in the equipment technical manual.

3.2.4 Attenuation versus ambient noise. The steps of announcing system attenuation versus ambient noise levels shall be as specified in table I. The ambient noise shall be measured as specified in ANSI S1.4 for sound level meter C response.

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3.2.4.1 Noise circuit level adjustment. A calibrated precision level continuous adjustment shall be provided for changing the maximum levels of operation of table I to any maximum level from 110 to 80 dBC.

3.2.5 General design requirements. NAVSEA technical manual SE101-AA-MMO-010 shall be applied to cover items not addressed in this specification. In the event of conflict, the requirements herein shall prevail. Failure detected by the BITE monitoring is required to be indicated by a flashing light (see 3.2.1.1). This light shall be on the front face cover. In addition, the operating mode selection switch shall also be on the front face cover for external access. This switch shall include the "OFF" selection (OFF-BYPASS-RUN). The test selection shall be internal.

3.2.6 Visual defects. Workmanship and visual inspections shall conform to MIL-E-16400 surface examination and workmanship requirements.

3.2.7 Audio power. The ALC announcing signal power servicing capability shall be equal to or exceed table III requirements.

TABLE III. Power requirements (see 3.2.7 and 3.4).

Characteristics	Requirements
(a) Input power	
1. Central amplifier/SIC	115 Vac, 60 Hz, single phase, type I in accordance with DOD-STD-1399, Section 300
2. Distributed amplification/WIC	28 Vdc (14 to 35 Vdc)
(b) Audio power	Minimum:
1. Central amplifier (70.7 volt)	40 watts rms continuously duty-cycled 1 minute on and 4 minutes off (125 ohm full load)
2. Distributed amplification (12.2 volt)	60 milliwatts continuous

3.3 Mounting, dimensions, and clearances. The ALC shall be bulkhead mounted. The mounting footprint maximum dimensions shall be 13.25 by 13.25 inches including cable stuffing tubes. The overall depth shall not exceed 6.6 inches. The front face clearance required shall be less than 13 inches. The microphone and vibration isolation mount assembly shall provide for bulkhead and for ceiling mounting and allow clearance for each orientation as necessary.

3.4 Input power. Service power requirements shall be in accordance with MIL-E-16400 and table III herein.

3.5 Physical design. Compliance with table IV shall be required.

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TABLE IV. Design and construction.

Characteristic	Requirement
Accessibility	MIL-STD-454, requirement 36
Marking	MIL-STD-454, requirement 67
Internal wiring practices	MIL-STD-454, requirement 69
Moisture pockets	MIL-STD-454, requirement 31
Metals and alloys	MIL-E-16400
Soldering	MIL-STD-454, requirement 5
Interchangeability	MIL-STD-454, requirement 7
Workmanship	MIL-STD-454, requirement 9
Safety	MIL-STD-454, requirement 1

3.6 Reliability requirements. The mean-time-between-failure (MTBF) requirements for the equipment, as defined herein, shall be more that 20,000 hours. MTBF prediction shall include all parts and shall be performed for the Naval unsheltered environment, in accordance with MIL-STD-785, task 203.

3.6.1 Parts selection. Electrical/electronic parts shall meet the following minimum requirements:

(a) Microelectronic parts shall meet the requirements of MIL-M-38510, class B.
(b) Discrete semiconductor parts shall meet the requirements of MIL-S-19500.
(c) Electrical or electronic parts shall meet the requirements of reliability of 0.1 percent failure/1000 hours (P level quality) in accordance with NAVSEA 0967-LP-597-1011.
(d) Microelectronic parts other than those in (a) above, which have been screened, burned-in, and tested to the class B product assurance requirements of MIL-M-38510, may be used in the equipment only if approved by the contracting activity.
(e) Discrete semiconductor parts other than those in (b) above, which have been screened, burned-in, and tested to the JANTX quality assurance level of MIL-S-19500, may be used in the equipment only if approved by the contracting activity.

3.6.2 Parts derating. Electrical/electronic parts shall not be stressed greater than the percentages of rated values specified in table V.

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TABLE V. Electronic parts derating.

Part type	Derating parameters	Derated to percent of rated [1] [2]
Resistor:		
Carbon composition	Power/voltage	50/80
Film high stability metal	Power/voltage	40/80
Wirewound accurate	Power/voltage	40/80
Wirewound power	Power/voltage	40/80
Wirewound chassis mounting	Power/voltage	40/80
Variable wirewound	Power/voltage	40/80
	current	70
Variable non-wirewound	Power/voltage	50/80
	current	70
Thermistor	Power/voltage	50*/80*
*Usage/accuracy dependent		
Tantalum nitride chip	Power/voltage	50/80
Capacitors:		
All	Observe ripple voltage rating	
Ceramic	Voltage	50
Glass	Voltage	50
Mica	Voltage	60 dipped, 40 molded
Plastic	Voltage	50
Mylar	Voltage	60
Paper	Voltage	50
Tantalum solid	Voltage	50
	Reverse voltage	2
	Circuit impedance	greater than 3 ohms per volt
Tantalum wet	Voltage	60
	Reverse voltage	0
Tantalum foil	Voltage	60
Aluminum electrolytic	Voltage	55 to 80
Relays:		
All	Use arc suppression	
	Contact current (continous)	50 resistive 25 inductive
	Contact current (surge)	80
	Coil engerize voltage	Hold to mfg. nominal
	Coil dropout voltage - vibration	Rating 75 (including "Q" of mounting)
	Contact gap	.005 minimum opening

See footnotes at end of table.

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TABLE V. Electronic parts derating. - Continued

Part type	Derating parameters	Derated to percent of rated [1] [2]
Switches	Contact current	30
	Voltage	50
Connectors:		
	Contact current	50
	Voltage (dielectric withstanding)	25
Magnetic devices:		
	Power	50
	Current density	2.0 mA per circ. mil
	Current (continuous)	60
	Current (surge)	90
	Voltage (continuous)	60
	Voltage (surge)	90
	Hot spot temperature (operating)	75
	Insulation breakdown Voltage	25
RF coils:		
	Current	50
Transistors:		
Bipolar, power and FET	Power	40
	Forward current (continuous)	50
	Forward current (surge)	60
	Voltage (steady state)	75 across any junction
	Transient peak voltage	80 across any junction
	Reverse junction voltage	65 across any junction
	Junction temperature	100°C maximum
Diodes:		
Switching, general purpose	Current (surge)	70
Rectifier, general	Current (continuous)	60
	Power	50
	Peak inverse voltage	65

See footnotes at end of table.

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TABLE V. Electronic parts derating. - Continued

Part type	Derating parameters	Derated to percent of rated [1] [2]
	Current (surge)	70
	Current (continous)	60
	Power	50
SCR	Current (surge)	70
	Current (continuous)	60
	Peak inverse voltage	65
All	Junction temperature	100°C maximum
Microcircuits:		
All	Combination of ac and dc loads is not recommended	
Linear	Current (continuous)	70
	Current (surge)	60
	Voltage (signal)	75
	Voltage (surge)	80
	Voltage reverse Junction (signal)	65
	Voltage reverse Junction (surge)	85
	Junction temperature	100°C maximum
Digital	Supply voltage	Hold to MIL-M-38510 nominal ratings and tolerances
	Junction temperature	100°C maximum
	Fanout	80
Crystals:		
	Drive voltage	50
Fuses:	Current	75

[1] The maximum ratings as determined by the Safe Operating Areas (SOA) curves for power switching transistors shall not be exceeded.

[2] Many families of digital microcircuits exhibit additional characteristics which may require derating (for example, toggle frequency, hold times, and so forth).

3.7 Maintainability requirements. The circuit card assembly and power supply shall be removable using only a screwdriver. The mean-time-to-repair (MTTR) the ALC shall be not more than 15 minutes. The maximum repair time shall be 1 hour at the 95th percentile. Accessibility and marking shall be as specified in table IV.

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3.8 Environmental. The ALC shall meet performance requirements (see 3.2 and 3.4) during and after the environmental conditions specified in table VI except during shock.

TABLE VI. Environmental conditions.

Condition	Requirement
Temperature	MIL-E-16400, range 4 (modified 0-65°C)
Humidity	MIL-E-16400
Vibration	MIL-STD-167-1, type I
Shock	MIL-S-901, grade A, type A, class I
Enclosure	MIL-STD-108, dripproof (45 degrees)
Salt spray	MIL-E-16400 (sheltered equipment)
Electrical power	MIL-E-16400 voltage and frequency regulation range

3.9 Electrostatic discharge protection requirements. If parts sensitive to electrostatic discharge (ESD) are used in the equipment, the contractor shall:

(a) Incorporate protective circuitry in equipment design.
(b) Ensure that ESD sensitive parts and subassemblies are protected in all phases of handling and testing.
(c) Affix warning labels to protective packaging and to the equipment.
(d) Provide warnings in all relevant areas of the equipment technical manual.
(e) Provide identification marking on all ESD sensitive subassemblies visible to maintenance personnel (that is, on card edge) prior to maintenance handling of the equipment.
(f) Ensure that spare parts, modules, printed circuit boards, subassemblies, and so forth are protected from ESD damage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with

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all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examinations of 4.6 and tests of 4.7.

4.3.1 Test report. The contractor shall prepare a first article test report in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.4 Quality conformance inspection. Quality conformance inspection shall be conducted on all equipment selected in accordance with 4.5 and shall consist of the examination and tests specified in table VII.

TABLE VII. First article and quality conformance inspection.

Inspection	Requirement	Test	First article	Quality conformance
Visual examination	3.2.6 and 3.5	4.6	X	X
Burn-in	3.2.8	4.7.1	X	X
Built-in test equipment	3.2.1.2	4.7.2.1	X	X
Signal circuits	3.2.1.1	4.7.2.3	X	X
Noise circuits	3.2.3	4.7.2.4	X	X
Input power	3.4	4.7.2.5	X	
Audio power	3.2.7	4.7.2.2	X	
Maintainability	3.7	4.7.3	X	
Environmental	3.8	4.7.4	X	

4.5 Sampling. Sampling shall be conducted as required by the contract (see 6.2.1) for the reliability test. Tests shall be in accordance with 4.7.3 on a sample having passed table VII quality conformance tests.

4.6 Examination. Examination shall be in accordance with MIL-E-16400, see 4.7.4.

4.7 Tests. The signal shall be a 620 hertz (Hz) tone at 70.7 volts root mean square (Vrms) for the central amplifier/SIC and at 12.2 Vrms for the distributed amplifier/WIC announcing systems. The maximum noise shall be 99 dBA (voltage equivalent).

4.7.1 Burn-in. The ALC shall be turned on and placed in the "test" mode. A reference signal (0 dB) shall be applied to the signal input and no noise applied to the noise input. The BITE shall be used for input.

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4.7.2 Performance. Circuit component designators such as DS3, S-2, and so forth shall be in accordance with technical manual SE101-AA-MMO-010.

4.7.2.1 BITE. The ALC shall be turned on and placed in the "test" mode. DS6 shall light, and one of the switch level indicators (DS1 through DS5) shall light. The switching characteristics shall be tested as follows:

Cycle switch S-2 through positions 1 to 11, allowing at least 6 seconds in each position before switching to the next position. The following indications shall be given for each switch position:

S-2 position	Indication
1	DS1
2	DS2
3	DS2
4	DS3
5	DS3
6	DS4
7	DS4
8	DS5
9	DS5
10	Sequentially stepped from DS5 to DS1 (15 sec max for C-10756/SIC) (60 sec max for C-10757/WIC)
11	Sequentially stepped from DS1 to DS5 (45 sec max)

Move switch S-1 to the "self-test" position; DS7 shall light. Place the ACL in the "run" mode; DS-6 shall light after 8 to 12 seconds. Any improper indication shall be classified as a minor failure.

4.7.2.1.1 Microphone. A microphone input test shall be conducted as follows: Connect the microphone to the noise input. Turn the ALC on and place it in the "test" mode. Move switch S-1 to the "self-test" position; DS7 shall light. Allow switch S-1 to return to neutral; DS7 shall go out. Disconnect the microphone and move switch S-1 to the MIC-TEST position; DS7 shall not light. An improper indication shall be classified as a major failure.

4.7.2.1.2 Noise circuit monitor test. Verify conformance to 3.2.1.1.1 BITE monitoring circuit requirements by making the following failures occur:

(a) Microphone diaphragm mechanically restricted.
(b) Microphone line open.
(c) Microphone line shorted.
(d) ALC noise amplifier signal path open.
(e) ALC noise circuitry power failure.

Report function of signal output attenuator and failure light and recovery from failed state.

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4.7.2.2 Audio output power test.

4.7.2.2.1 Central amplifier type/SIC. Connect a 40 watt or more loudspeaker load. Provide a "Rated" (70.7 volt) 1000 Hz signal for 1 minute ON and 3 minutes OFF continuously for 1 hour after the ALC chassis temperature has stabilized in ambient air temperature of 65°C. Demonstrate that ALC operation is normal during each 1 minute ON time by continuous recordings.

4.7.2.2.2 Distributed amplifier type/WIC. Connect a 60 or more milliwatt resistive load. Provide a "Rated" (12.2 volt) 1000 Hz signal as specified in 4.7.2.2.1 and record same data.

4.7.2.3 Signal sensing.

4.7.2.3.1 Signal onset time. A reference signal shall be applied to the signal input with no noise input. Signal onset time shall be the time required for the output signal to drop from 0 dB to minus 26.5 dB. DS6 shall light. The result shall conform to table II.

4.7.2.3.2 Signal release time. Signal release time shall be the time required for DS6 to go out after removal of the input signal. The result shall conform to table II.

4.7.2.3.3 Minimum signal for activation. A signal of not more than 0.5 Vrms shall be applied to the signal input. The voltage of the signal shall be slowly increased until DS6 lights. The minimum signal for activation shall be the minimum signal at which DS6 lights. The result shall conform to table II.

4.7.2.4 Noise sensing. Noise sensing shall conform to the requirements of 3.2.1.1, 3.2.4, and tables I and II herein.

4.7.2.4.1 Noise attack time. The time shall be measured from "step" (10 milliseconds (ms) or less) noise level increase to attenuator change completion. The time shall be measured for a plus 18 dBA change, a minus 18 dBA change, and a plus 6 dBA change at 88 dBC.

4.7.2.4.2 Noise release time. Noise release time shall be the transition time from maximum volume to minimum volume. This time may be obtained using the built-in test (see 4.7.2.1) by measuring the time of sequential stepping from DS1 to DS5 when S-2 is switched from position 1 to position 11. The results shall conform to table II.

4.7.2.4.3 Noise switching. Noise levels at which switching occurs shall comply with table VIII as required by 3.2.4. Test for switching at the table VIII indicated rms sound pressure level.

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TABLE VIII. Indications for switching characteristics.

Ambient noise-sound pressure dBC		Lighted indicator	Signal attenuation re "rated"
Low range	High range		
> 90	> =100	DS1	0 dB
< 90	< 100	DS2	-6 dB
84	94	DS2	-6 dB
< 84	< 94	DS3	-12 dB
78	88	DS3	-12 dB
< 78	< 88	DS4	-18 dB
72	82	DS4	-18 dB
< 72	< 82	DS5	-26.5 dB

4.7.2.5 Electrical input power. Electrical input power characteristics and insulation resistance shall be tested in accordance with table III and MIL-E-16400 electrical tests. The results shall conform to table III and MIL-E-16400.

4.7.3 Maintainability. The equipment shall undergo a maintainability demonstration in accordance with MIL-STD-471, test method 9. The test shall demonstrate the equipment design meets organizational level corrective maintenance, fault detection, localization, and isolation requirements. The following conditions shall also govern the demonstration:

(a) The tasks shall be selected in accordance with MIL-STD-471, appendix A.
(b) The tasks shall be performed using the equipment recommended in the Navy accepted technical manual.
(c) Corrective maintenance; 20 tasks, using repetition times if sufficient tasks are not available.

If the demonstration detects equipment, software, or support item deficiencies or results in a reject decision, retest shall be required. Total or partial retest shall be as determined by the contracting activity. The tasks to be performed during demonstration shall include the following as a minimum:

(a) Remove and replace a circuit card assembly and run the built-in test.
(b) Remove and replace the power supply.
(c) Disconnect power, audio input, microphone input, and audio output; reconnect all inputs and outputs.

All tasks shall be started and counted complete with the ALC closed and operating normally, as verified using BITE. Each task shall be performed three times to obtain sample maintenance times for the calculation of MTTR.

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4.7.4 Environmental. Environmental test for temperature, humidity, and salt fog shall be performed in accordance with MIL-E-16400. The humidity test shall include an additional exposure at 95 percent relative humidity at 30°C for 24 hours with the ALC enclosure open and the ALC in "test" mode. Vibration shall be tested in accordance with MIL-STD-167-1. Shock shall be tested in accordance with MIL-S-901. The built-in test equipment and the 4.7.2.1 BITE test shall be used to verify ALC performance during exposure and after exposure.

4.7.4.1 Random vibration screening. Each equipment shall be subjected to the following vibration for the purpose of screening or finding assembly and electrical defects:

Vibration frequencies - 33 Hz, 55 Hz, 220 Hz for each direction.
Vibration duration - At least 3 minutes at each frequency in each direction.
Vibration amplitude - 0.04 G^2/Hz; G = 32 feet per second.
Vibration directions - In each of three orthogonal directions to the principle electronic circuit board plane and its connector direction of insertion.
Electrical operational test (after vibration) - Using the BITE, observe that the level switching functions correctly for each noise level.

The item may be corrected and retested. Redesign is required if on retest the item fails the same way again.

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 Preservation, packing, and marking. ALCs shall be individually preserved level A, C, or commercial; packed level A, B, C, or commercial as specified (see 6.2.1); and marked in accordance with MIL-E-17555.

6. NOTES

6.1 Intended use. The automatic loudspeaker volume control is intended for use in announcing systems located in areas where the background noise levels change due to such things as machinery, wind, flooding and so forth, ambient noise level ranges that may be compensated for are defined by table I and 3.2.4.1.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of ALC (see 1.2).
(c) When first article inspection is required (see 3.1).
(d) Sampling required (see 4.5).
(e) Levels of preservation and packing or commercial requirements (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraph.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.3.1	Test report (first article)	DI-T-2072	10.1.b

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection of those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.4 Definitions. The definitions given in 6.4.1 through 6.4.4 apply to automatic loudspeaker controls.

6.4.1 Signal onset time. Signal onset time is the time from minimum signal for activation to noise level latch and attenuator switch completion (see table II).

6.4.2 Noise attack time. Noise attack time is the time required to reach the value of maximum noise level (the level that would result in zero attenuation) from the minimum level (the level that would result in maximum attenuation) (see table II).

6.4.3 Noise release time. Noise release time is the time required to reach the value of minimum noise level (the level that would result in maximum attenuation) from the maximum noise level (level that results in zero attenuation) (see table II).

6.4.4 Signal release time. Signal release time is the time from below minimum signal for activation to release deactivation (see table II).

6.5 Subject term (key word) listing.

Amplifier
Announcing systems
Audio
Loudspeaker
Microphone
Noise input
Visual defects

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
(Project 5830-N067)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-L-24681(SH)	LOUDSPEAKER, CONTROL OF VOLUME, AUTOMATIC (METRIC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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